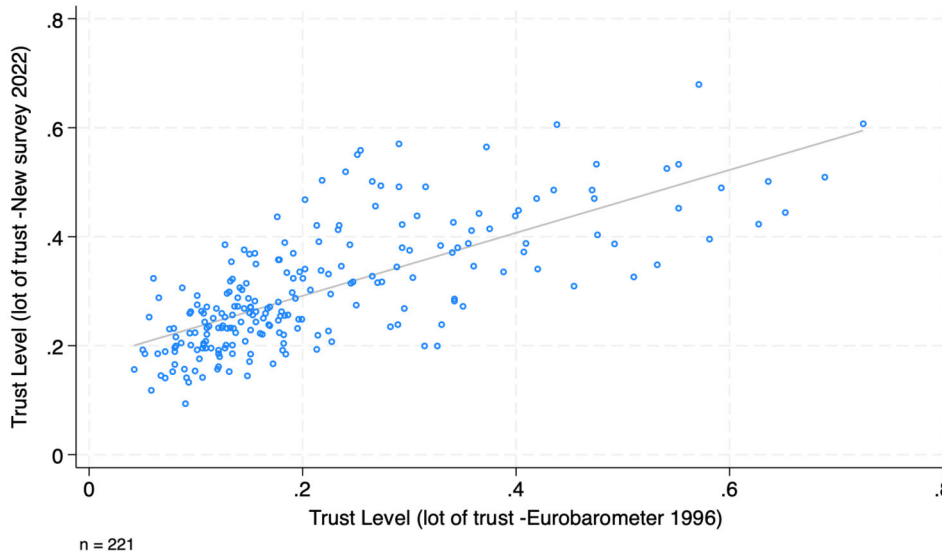
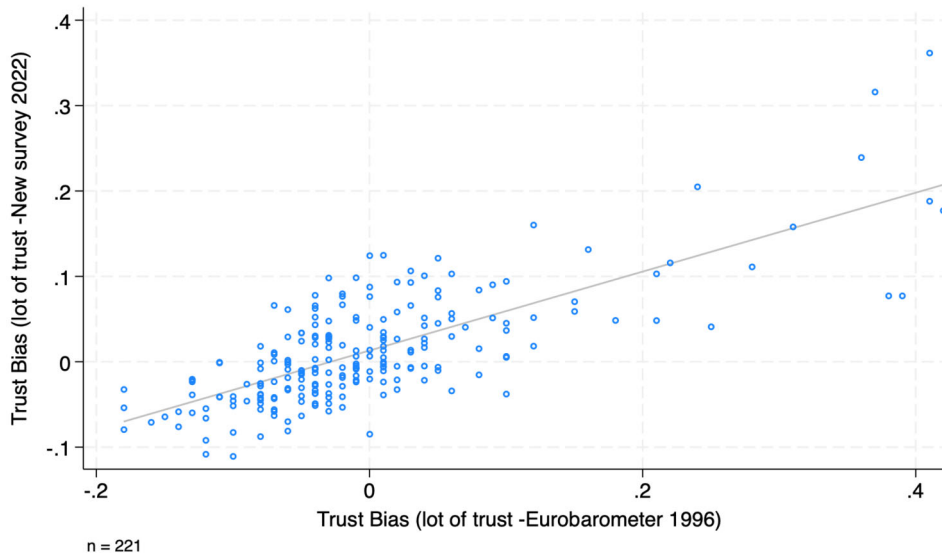


(a) Trust levels



(b) Trust biases

FIGURE 4. Trust levels and biases (lot of trust)—Eurobarometer vs. new online survey. This figure represents the scatterplot of the trust levels and biases as measured via Eurobarometer (*x*-axis) and our new online survey (*y*-axis). Trust in both cases is defined as the portion of individuals in home country expressing "a lot of trust" toward target country. Unit of observation is at home–target-country level and sample size is 221.

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the strongly positive relationship between the two surveys conducted more than a quarter century apart, consistent with the notion that cultural biases are persistent. The correlation coefficients between the two surveys are 0.72 for trust levels and 0.75 for trust biases (both significant at the 0.01 level).[47]

Next, we check the external validity of our main results using this new survey covering 30 countries (as opposed to 15 in Eurobarometer). This not only gives us a chance to include banks headquartered in additional European countries but also expands the target countries in our sample.[48] Panel A of Table 7 employs the same definition of trust (based on responses affirming "a lot of trust") as in Table 2, while panel B does the same for the alternative measure based on grading responses from 1 to 4.[49] The coefficients of interest are all statistically significant at the 5% level or better. Estimates in panel A for bank-level trust bias are strikingly similar in size to our baseline estimates in Table 2 (especially in column (5)).[50] The results thus support the external validity of our findings and confirm our assumption that cultural stereotypes are persistent over time.

We additionally check the externality validity of our findings vis-à-vis other types of exposures that banks have in the same target countries. For this purpose, we separately collect data from EBA on banks' corporate exposures, which cover only a subset of banks and disclosure dates. Table 8 presents the results, where the dependent variable is defined as a dummy indicating positive corporate exposures in the target country and, alternatively, a continuous variable with log nominal values (in millions) of corporate exposures. In line with the findings of Giannetti and Yafeh (2012) and Hagendorff et al. (2023), we detect a positive and significant relationship between cultural trust and banks' corporate exposures.[51]

47. These high correlations exist despite certain structural differences between our new survey and Eurobarometer. First, our survey was undertaken online, whereas Eurobarometer takes place in person, leading to different selection biases in sampling. Second, our survey has smaller sample sizes per country. Third, different demographic characteristics have been used to make the two surveys nationally representative.

48. A caveat is that our variable of interest (i.e., trust) is in this case measured after sovereign exposures are realized and thus could suffer from endogeneity. The fact that our sample period includes sovereign debt crises heightens the possibility of reverse causality from sovereign debt exposures to trust (measured later in time). Hence, we utilize the historical Eurobarometer surveys for the baseline and additional results in the paper.

49. This is the same definition used in Online Appendix Table C.10.

50. Similarly, the estimates in panel B are very similar to those in Online Appendix Table C.10. However, despite the increase in sample size, standard errors are now substantially larger, potentially reflecting the smaller samples of individuals per country in our new survey compared to Eurobarometer. Online Appendix Table D.4 reports corresponding results when dropping all domestic observations from the sample. Estimates are again comparable but are not as statistically precise as before, especially in more saturated specifications.

51. Results are very similar, even when we focus only on foreign exposures of these banks (see Online Appendix Table D.5).

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TABLE 7. External validity—bank-level trust bias (from online survey) and probability of sovereign exposure.

Outcome →	(1) Sovereign exposure	(2) Sovereign exposure	(3) Sovereign exposure	(4) Sovereign exposure	(5) Sovereign exposure
Panel A					
Bank-level trust bias (online—lot of trust)	1.439***	1.531***	1.549***	1.508***	1.561***
	[0.176]	[0.428]	[0.427]	[0.419]	[0.470]
Panel B					
Bank-level trust bias (online—graded)	0.486***	0.664***	0.679***	0.664***	0.682**
	[0.075]	[0.222]	[0.223]	[0.224]	[0.266]
Control for bank-level branches	No	No	Yes	Yes	Yes
Control for bank-level branches (squared)	No	No	No	Yes	Yes
Bank × time FEs	Yes	Yes	Yes	Yes	Yes
Target-country × time FEs	Yes	Yes	Yes	Yes	No
Home-country × target-country FEs	No	Yes	Yes	Yes	No
Home-country × target-country × time FEs	No	No	No	No	Yes
Observations	47,520	47,520	47,520	47,520	43,230

Notes: The table summarizes the results of equation (1) estimated over the full sample period from 2010-Q1 to 2021-Q2. Panel A and panel B display separate estimations with two different independent variables, both of which are sourced from our new online survey undertaken across 30 European countries in the second half of the year 2022. Dependent variable is *Sovereign Exposure*, defined as a dummy variable indicating any positive exposure of a bank toward a target country at a point in time reported in EBA and CEBS disclosures. *Bank-level Trust Bias (online—lot of trust)* is computed for each bank–target-country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the portion of individuals in home country expressing "a lot of trust" toward target country. *Bank-level Trust Bias (online—graded)* is computed for each bank–target-country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the average across individuals in home country expressing values from 1 (i.e., "no trust at all") to 4 (i.e., "lot of trust") toward target country. *Bank Branches* measures the number of bank branches (in thousands) that the bank owns in the target country. All branch-related information is from SNL Financial. For the detailed construction of the data, see Section 3. Robust standard errors are clustered at the bank level and reported in brackets. *Significant at 10%; **significant at 5%; ***significant at 1%.

5.5. *Heterogeneity*

Figure 5 reports additional results for banks with more and fewer total assets, banks with more or less widely diversified bond portfolios, and target countries more and less frequently present in those portfolios. In panel A, we categorize banks based on their average total assets within our sample period. In panel B, we compute the number of countries to which a bank has positive exposures and average it over time for each bank. This allows us to calculate a time-invariant measure of diversification and to separate high- and low-diversification banks by choosing the median bank as a threshold. In panel C, we compute the number of times the bonds of a target country are included in portfolios across all banks and times, and then separate countries into two groups based on median values.

Panel A confirms that stereotypes influence the sovereign portfolios of both large and small banks. Panel B shows that banks whose investment portfolios are

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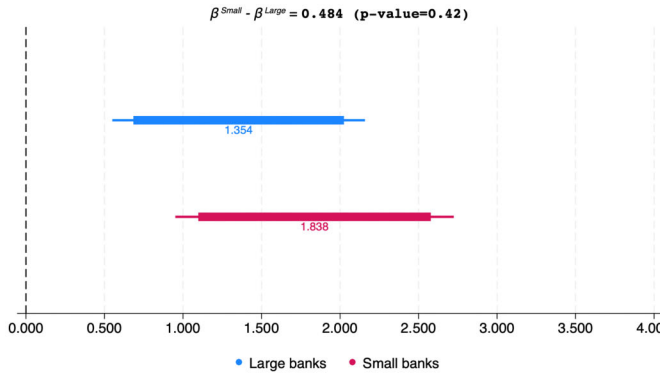
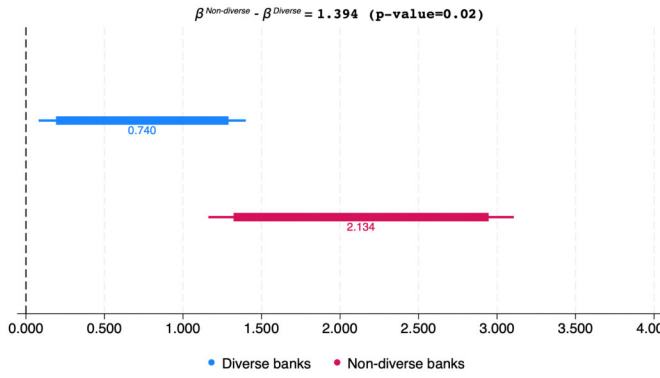
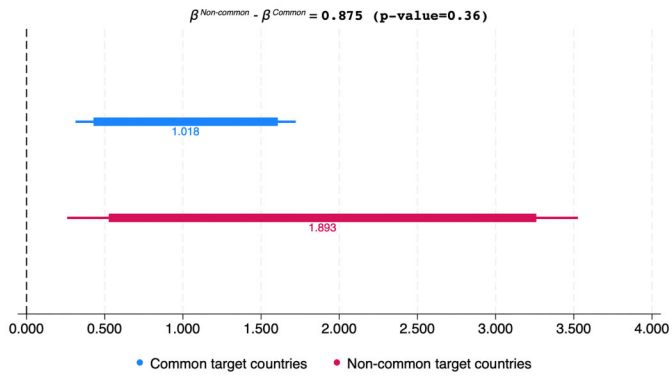


(a) Bank size



(b) Bank diversification



(c) Target country commonness

FIGURE 5. Bank- and country-level heterogeneity in baseline results. The figure summarizes the baseline results estimated for six different subsamples over the full sample period from 2010-Q1 to 2021-Q2. The estimated specification corresponds to column (5) in Table 2, and the coefficients for *Bank-level Trust Bias* are plotted for each subsample. Thicker lines indicate 90% and thinner lines indicate 95% confidence intervals.

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